

HLV as a possible subset of FFGFT

Candidate note: embedding map via the Hilbert-space transformation

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Starting point

In the course of the bridge analysis between FFGFT and HLV, a finding emerged that is recorded here as a directed candidate note. It is not a closed derivation; it names the next clean step (P35).

Three steps of retrospect

Step 1 — original blockade. The C_3 -in- A_5 geometry of the HLV framework (icosahedral projector, $\tau = \phi$, $\theta = 2/9$ as output) was not directly visible on $T^4/\mathbb{Z}_3$. A direct geometric identification was blocked: T^4 is compact 4-dimensional, A_5 is the symmetry group of the icosahedron in 3D — the dimensions do not match straightforwardly.

Step 2 — Hilbert-space transformation opens the path. The lossless bijection $H = L^2(T^4) \otimes \mathbb{C}^3$ (Doc. 230/282) translates $T^4/\mathbb{Z}_3$ into an operator algebra on the $\mathbb{C}^3$ fibre. In $\mathbb{C}^3$, $C_3 < A_5$ is naturally accessible: the $\mathbb{Z}_3$ circulant diagonalisation on the fibre outputs $\theta = 2/9$ as a phase (Doc. 291), and the C_3 -in- A_5 geometry is demonstrable as a parameter-free invariant on the same fibre (Doc. 293). The transformation made visible what was geometrically hidden directly.

Step 3 — consequence. If FFGFT contains $C_3 < A_5$ via the Hilbert-space transformation, and HLV is built on $C_3 < A_5$ with $\tau = \phi$, then HLV's geometric core could be a *subset* of the FFGFT structure — not a peer framework, but a specific projection of FFGFT's $\mathbb{C}^3$ -fibre geometry.

What is established (from available documents)

- FFGFT contains $C_3 < A_5$ and $\theta = 2/9$ via the Hilbert-space transformation — documented in Doc. 291 (circulant phase) and Doc. 293 (parameter-free C_3 -in- A_5 invariant, second witness).
- HLV's geometric object (the ϕ -icosahedral projector) carries the same $C_3 < A_5$ — one-sidedly verified (FFGFT $\rightarrow$ HLV, Doc. 294).
- The verification direction is asymmetric: FFGFT derives $\theta = 2/9$; HLV books it as a candidate overlap without its own derivation — confirmed by inspection of his available documents (Stage 19 manuscript, HLV-Consciousness-Bridge1, Runs H–J): $\theta = 2/9$ appears in none of HLV's internal derivations.
- His H_2 operator (Run H, skew_J) contains $1/\phi$ and $1/\phi^2$, but no C_3 -in- A_5 output and no $\theta = 2/9$ as a result. Run H is moreover *negative/underdetermined* (does not survive the u -shuffle specificity test).

What is not shown (open obligation)

That HLV's ϕ -icosahedral layer is *fully* contained in the FFGFT $\mathbb{C}^3$ fibre requires an explicit *embedding map*

$$\iota: \text{HLV ground object} \hookrightarrow L^2(T^4) \otimes \mathbb{C}^3, \quad (1)$$

which has not yet been constructed. "Subset" is the structurally plausible conjecture — not a proven theorem.

Next clean step

Construct or falsify the embedding map ι :

1. Show that HLV's ϕ -icosahedral projector is representable as an operator on FFGFT's $\mathbb{C}^3$ fibre — i.e. as an element of $\text{End}(\mathbb{C}^3)$ with C_3 invariance.
2. Check whether HLV's spectral data (Run J, native-6D candidate) are compatible with the discrete spectrum of the T^4 connection-Laplacian (Doc. 283) — or whether they deviate and thus block the embedding.
3. If ι exists: HLV is a specific projection of FFGFT — its geometric layer is a subset, while its dynamical objects (Spiral-Time, U_3) remain independent and unembedded.
4. If ι cannot be constructed: both frameworks share a common object ($C_3 < A_5$, $\theta = 2/9$) but are structurally independent — no subset relation.

Concrete embedding candidate: the Markov limit and below

The question of whether Marcel's Archimedean spiral time finds a structurally defined place in FFGFT can be answered precisely — not as an analogy but as a limit. And the analysis reveals that the object is even weaker than initially assumed.

Archimedean = Markov limit — but U_3 is weaker still. The FFGFT memory kernel (Fourier transform of the discrete spectral density $\sum_k g_k^2 \delta(\omega - \omega_k)$, Doc. 283) becomes Archimedean exactly when the couplings $g_k \rightarrow 0$ — the *Markov limit* of open quantum systems. In this limit the kernel still accumulates without bound on $\mathbb{R}$. Marcel's U_3 (Eq. 7 of his Consciousness document) goes further: the window M_m is finite and fixed, older values actively drop out (*sliding window*). This yields a **hard-reset kernel** — a memory kernel with finite support:

$$K(t - \tau) = 0 \quad \text{for } t - \tau > M_m \cdot \Delta t. \quad (2)$$

This is not merely Archimedean — it is *periodically bounded*. U_3 does not accumulate without bound; it closes after M_m steps.

Does the window end before the first closure period? In FFGFT's frozen configuration (Doc. 295), the winding closes after exactly 75 turns ($1/(100\xi_0) = 75$) — the smallest coherent unit of the recursion. If Marcel's window width $M_m < 75$ steps (in the winding coordinate), then U_3 does not even accumulate one complete closure period of 75 turns (according to Doc. 295 one step is one turn and 75 is the number of turns until closure). There is no closed winding, no geometric spiral — only a sliding memory window that renews with each step. The object lies **below the smallest coherent unit** of the FFGFT recursion.

Refined embedding candidate. The concrete location of U_3 in FFGFT is therefore not the Markov limit (which still accumulates on $\mathbb{R}$) but the *hard-reset kernel below one closure period*:

$$\iota: U_3 \hookrightarrow \{K \in L^2(T^4) \otimes \mathbb{C}^3 \mid K(t - \tau) = 0 \text{ for } t - \tau > M_m \cdot \Delta t, M_m < 75\}. \quad (3)$$

This is present in FFGFT — but as the **degenerate boundary case**, not as a regular configuration. The hierarchy is therefore clear:

$$U_3 \subset \text{Markov limit} \subset \text{Archimedean} \subset \text{log spiral } (e) = \text{FFGFT kernel (Doc. 295/283)}. \quad (4)$$

Each level is more information-poor than the next. U_3 is the poorest.

Falsification condition

The embedding ι is falsifiable: if Marcel's U_3 carries *more* in his data than the hard-reset kernel — i.e. if non-Markov signatures (revivals, BLP backflow, oscillatory kernel, or correlations beyond M_m steps) are detectable — then U_3 exceeds the boundary case and a deeper embedding is needed.

Marcel's own discipline (U3 ablation test, Table 4 of his Consciousness document) is exactly the right test: if U_3 does not exceed the Markov baseline, the embedding in the boundary case is correct; if it does, a richer layer is needed. The programme is clean — he only needs to run it to the end.

Two HLV spiral times — two different embedding candidates

Doc. 295 §9 records a distinction that is also load-bearing for the embedding question of Doc. 297: "Spiral time" in HLV denotes not one object but two.

First object — the geometric time from the factorisation $T^7 = T^4 \times T^3$ (Doc. 285): an A_5 singlet on the S^1 factor, decoupled and separable. This limit corresponds in the K_{frak} recursion to the fixed point $\xi \rightarrow 0$ — the defect vanishes, the winding closes after one turn (according to Doc. 295; the 75-closure belongs to frozen ξ_0 with $d = 1/75$), the pitch becomes constant. This is the *geometric* boundary of the FFGFT spiral.

Second object — the temporal-dynamic spiral time, the object HLV *calls* by that name: the triadic operator $\Psi(t) = (t, \phi(t), \chi(t))$ with a memory channel, without A_5 and without an S^1 factor. This operator is introduced in Krueger et al. (2026, doi:10.47852/bonviewMEDIN62029043) as a deterministic phase-memory decomposition and implemented as a numerical operator in the HLV G-Lattice Runs H-J (Zenodo, DOI not yet published). In the Consciousness document (Krueger, 2026, v0.2, not yet on Zenodo) it appears as $\Psi(t) = (U_1, U_2, U_3)$ with the explicit implementation Eq. 7 — a bounded window accumulator.

Two different embedding candidates in FFGFT:

The *geometric* object (A_5 singlet, S^1 , decoupled) finds its natural place at the $\xi \rightarrow 0$ fixed point of the recursion — the geometrically separable, information-poorest configuration of compact time. It is geometrically richer than U_3 , but carries no ratio e (Doc. 295 §10: blocked on the e -criterion).

The *temporal-dynamic* object (U_3 , hard-reset kernel, $M_m < 75$) finds its natural place in the Markov limit of the FFGFT memory kernel (Doc. 283, $g_k \rightarrow 0$) — below the smallest coherent unit of the recursion. It is the more information-poor of the two.

The hierarchy applies to the temporal-dynamic object:

$$U_3 \subset \text{Markov limit} \subset \text{Archimedean} \subset \log \text{ spiral } (e) = \text{FFGFT kernel}.$$

The geometric singlet lies to the side of this hierarchy — it is Archimedean ($\xi \rightarrow 0$, constant pitch), but from geometric factorisation, not from dynamic accumulation. Both boundary cases meet in the Archimedean regime, but they are different paths to it.

Regardless of whether U_3 is represented linearly, in polar coordinates, or as a spiral, the structure of the dependence remains the same: finite short-term memory with a fixed horizon M_m , without accumulation. The spiral representation is a *choice of coordinates*, not a structural property of the object — in every representation U_3 remains a hard-reset kernel with finite support $M_m < 75$. The geometric richness lies in the representation, not in the object. This is the same line as P35 ("translation is not justification") applied in the other direction: a richer representation does not make the object richer.

The deeper difference lies elsewhere: **accumulation is the actual memory achievement**. U_3 does not accumulate — it replaces. What was before M_m steps is irretrievably gone; no

past state acts on the future beyond the window. That is a short-term buffer, not memory in the structural sense.

The K_{frak} recursion is the opposite: $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ accumulates each step into the next — not as an explicit store, but as *structural imprinting*: the present state ξ_n carries the entire history of all preceding steps within itself. No step is forgotten, no window erases. The past is encoded in the object itself, not in a separate memory. This is memory in the deepest sense — not as storage but as a generative structure that carries its own history and from it produces the log spiral with ratio e (Doc. 295).

That is precisely what Marcel's consciousness framework requires (Section 1 of his Consciousness document: "memory-bearing, recursively integrated") — but what U_3 , by construction, does not deliver. The recursion would be the right carrier. U_3 is the degenerate boundary case below it.

The window as reduction — in the sense of Doc. 296

The window M_m in U_3 is itself an instance of the reduction principle of Doc. 296: a *projection onto a limited temporal segment*, just as a scale projection is a limited spatial segment. It leaves out (everything beyond M_m) and thereby makes a partial domain evaluable — the local, most recent dynamics. This is the same mechanism as in Doc. 296 (reduction as the condition of evaluability), only applied to the time axis rather than the scale axis. The window acts in the same sense as any projection.

The decisive difference therefore lies not in the fact that U_3 reduces — every evaluable description reduces — but in *whether the whole is named of which the reduction is a segment*.

FFGFT's reductions are exhibited as reductions of a known whole: the frozen 75-turn winding and the Markov limit of the memory kernel are segments of the full, accumulating, scale-invariant structure — one knows what has been left out and how to recover it (let the recursion run, do not freeze ξ , Doc. 295/283). The way back is laid out.

Marcel's window, by contrast, stands on its own, without the full object behind it — it is the projection without the projected. Recovery is not laid out because the full object is not named. The window evaluates the local temporal structure — a legitimate focus — but it is not recognisable as a segment of a boundaryless whole (Doc. 296: the embedded observer who sits on their scale without unrolling the recursion).

This is precisely what makes the embedding ι natural: it supplies the missing whole. U_3 as a reduction receives back, in FFGFT, the object of which it is a segment. The window acts in the same sense as any projection — the only difference is that FFGFT knows the way back and HLV does not (so far) lay it out.

Pre-registered prediction for the OP8-Neuro test (6 July 2026)

Marcel Krueger has published, with HLV-CONSCIOUSNESS-BRIDGE1 v0.3a (OP8-LOCK), a data-freeze / protocol-lock that fixes — before the run — a model ladder and a null family:

M_0	$\Delta\Phi$ / S-I-C only
M_1	U_1 only
M_2	$U_1 + U_2$
M_3	$U_1 + U_2 + U_3$
M_4	generic ML baseline

Central question (his formulation): whether the full $U_1/U_2/U_3$ model (M_3) provides measurable additional value over U_1/U_2 , information-only, coherence-only, **memory-free Markov**, phase-shuffled, time-shuffled, and the generic ML baseline — the U_3 *necessity test*.

This pre-registered test is simultaneously the decisive test for the embedding hypothesis of this document. FFGFT makes the following prediction about it — *before* Marcel's run, explicitly booked as a prediction (P35), not as a claim on his result:

1. **If U_3 is a hard-reset window with $M_m < 75$ (winding coordinate)** — i.e. a segment below one full closure period of 75 turns, the Markov limit of the embedding ι — then M_3 **does not significantly outperform the memory-free Markov baseline**. U_3 then lies entirely in the degenerate boundary case; the additional channel carries no structure beyond Markov.
2. **If M_3 clearly outperforms the memory-free Markov baseline**, then U_3 carries non-Markov structure (correlations beyond M_m , effective accumulation) that the hard-reset kernel does not explain. In that case the simple embedding ι into the Markov limit is *falsified*, and a richer embedding must be constructed.

The prediction is sharp because it hangs on Marcel's own construction: a bounded window M_m of fixed width (Eq. 7 of his Consciousness document, "avoids uncontrolled accumulation") can by construction carry no correlations beyond M_m — so the expected outcome is (1). An outcome (2) would indicate that U_3 , in implementation, does more than its definition suggests — which would be independently interesting and would then need to be localised.

This prediction does not touch Marcel's claim boundaries: it says nothing about consciousness, no physical validation, no HLV ontology. It concerns solely the structural question of whether U_3 carries beyond the Markov limit — and that is exactly what his OP8 test probes.

Status

Candidate note. Directed speculation on the basis of secured partial findings (P35). The subset conjecture is structurally plausible and falsifiable — that is its value. It is not a claim.